

Aim / Objective

To develop a Java wallet payment system using an interface, user-defined exceptions, encapsulation, and exception handling.

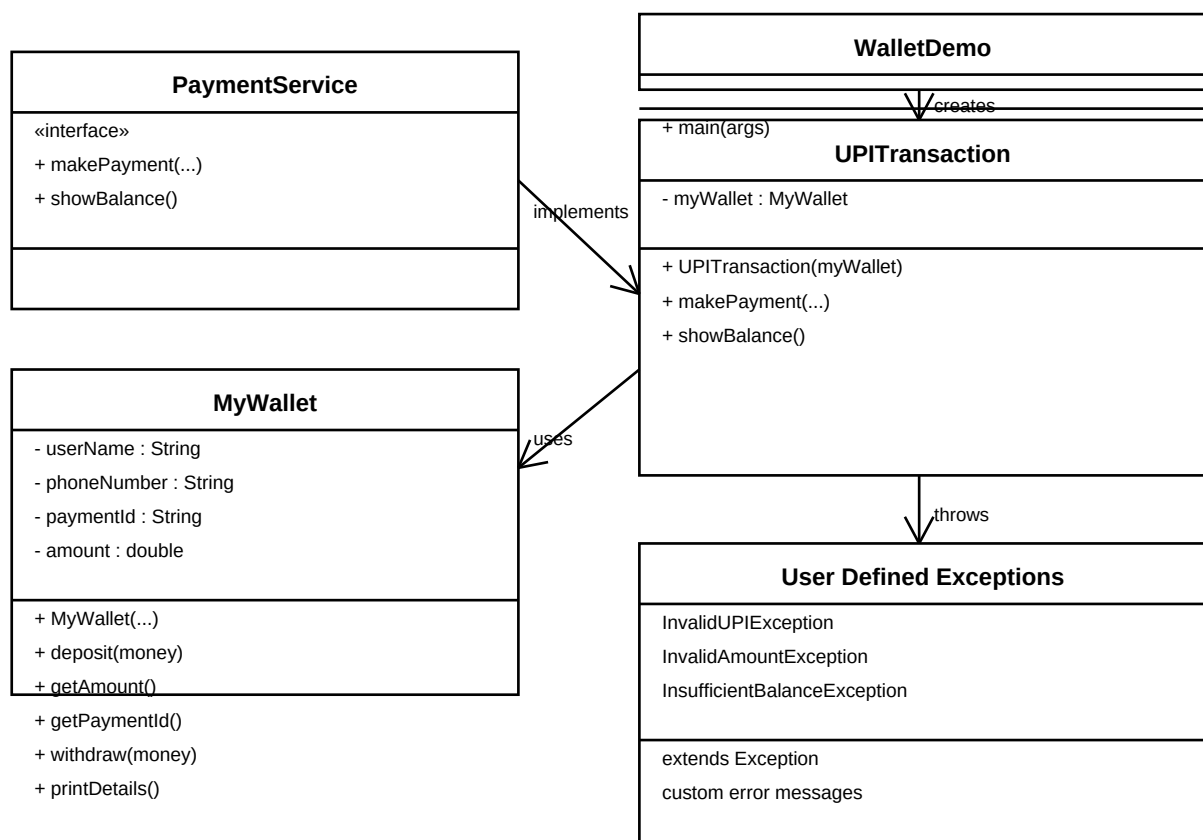
System Requirements

Java Development Kit (JDK) and any Java IDE such as Eclipse or IntelliJ IDEA.

Algorithm / Logic Description

1. Create the PaymentService interface with methods for making a payment and showing the wallet balance.
2. Create MyWallet to store user details and balance, with methods for deposit, withdrawal, and displaying details.
3. Create UPITransaction to implement PaymentService and connect the transaction with MyWallet.
4. Validate the UPI ID, payment amount, and available wallet balance before processing a payment.
5. Create user-defined exceptions for invalid UPI, invalid amount, and insufficient balance.
6. Handle the exceptions using try-catch and complete the transaction using finally.
7. Display the final wallet details after the transaction.

UML Class Diagram



Source Code

```
package walletsystm;
```

```

interface PaymentService {

    void makePayment(String upi, double money)
        throws InvalidUPIException,
               InvalidAmountException,
               InsufficientBalanceException;

    void showBalance();
}

// Wallet Account Class
class MyWallet {

    private String userName;
    private String phoneNumber;
    private String paymentId;
    private double amount;

    public MyWallet(String userName, String phoneNumber, String paymentId) {
        this.userName = userName;
        this.phoneNumber = phoneNumber;
        this.paymentId = paymentId;
        amount = 0.0;
    }

    public void deposit(double money) {
        if (money > 0) {
            amount = amount + money;
            System.out.println("Money Deposited: Rs." + money);
        } else {
            System.out.println("Invalid deposit amount.");
        }
    }

    public double getAmount() {
        return amount;
    }

    public String getPaymentId() {
        return paymentId;
    }

    public void withdraw(double money) {
        amount = amount - money;
    }

    public void printDetails() {
        System.out.println("\n----- My Wallet -----");
        System.out.println("User Name : " + userName);
        System.out.println("Mobile   : " + phoneNumber);
        System.out.println("UPI ID   : " + paymentId);
        System.out.println("Balance  : Rs." + amount);
    }
}

// Payment Class
class UPITransaction implements PaymentService {

    private MyWallet myWallet;

    public UPITransaction(MyWallet myWallet) {
        this.myWallet = myWallet;
    }

    @Override
    public void makePayment(String upi, double money)
        throws InvalidUPIException,
               InvalidAmountException,
               InsufficientBalanceException {

        // Checking UPI format
        if (upi == null || !upi.contains("@")) {

```

```

        throw new InvalidUPIException("Please enter a valid UPI ID.");
    }

    // Checking payment amount
    if (money <= 0) {
        throw new InvalidAmountException(
            "Payment amount should be greater than zero.");
    }

    // Checking wallet balance
    double remainingBalance = myWallet.getAmount() - money;

    if (remainingBalance < 0) {
        throw new InsufficientBalanceException(
            "Wallet does not have enough balance.");
    }

    myWallet.withdraw(money);

    System.out.println("\nPayment Done Successfully!");
    System.out.println("Amount Paid : Rs." + money);
    System.out.println("Receiver   : " + upi);
    System.out.println("Remaining   : Rs." + myWallet.getAmount());
}

@Override
public void showBalance() {
    System.out.println(
        "Current Wallet Balance: Rs." + myWallet.getAmount());
}
}

// User Defined Exceptions
class InsufficientBalanceException extends Exception {

    public InsufficientBalanceException(String msg) {
        super(msg);
    }
}

class InvalidUPIException extends Exception {

    public InvalidUPIException(String msg) {
        super(msg);
    }
}

class InvalidAmountException extends Exception {

    public InvalidAmountException(String msg) {
        super(msg);
    }
}

// Main Class
public class WalletDemo {

    public static void main(String[] args) {

        MyWallet account = new MyWallet(
            "Ravi Teja",
            "9876543210",
            "raviteja@upi");

        account.deposit(5000);

        UPITransaction transaction = new UPITransaction(account);

        try {

            transaction.showBalance();
            transaction.makePayment("friend@upi", 1500);

```

```

        transaction.showBalance();
    } catch (InvalidUPIException e) {

        System.out.println("Error: " + e.getMessage());
    } catch (InvalidAmountException e) {

        System.out.println("Error: " + e.getMessage());
    } catch (InsufficientBalanceException e) {

        System.out.println("Error: " + e.getMessage());
    } finally {

        System.out.println("\nTransaction Process Finished.");
    }

    account.printDetails();
}
}

```

Compilation and Execution Commands

```

javac WalletDemo.java
java WalletDemo

```

Output

```

PS C:\Users\Student> javac WalletDemo.java
PS C:\Users\Student> java WalletDemo
Money Deposited: Rs.5000.0
Current Wallet Balance: Rs.5000.0

Payment Done Successfully!
Amount Paid : Rs.1500.0
Receiver   : friend@upi
Remaining  : Rs.3500.0
Current Wallet Balance: Rs.3500.0

Transaction Process Finished.

----- My Wallet -----
User Name : Ravi Teja
Mobile    : 9876543210
UPI ID    : raviteja@upi
Balance   : Rs.3500.0
PS C:\Users\Student>

```

Result

The wallet payment system was implemented successfully using interface-based design, encapsulation, user-defined exceptions, and exception handling.